

Q1. What's the difference between a normal linear regulator and a switching regulator? Which one is better?

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A1. **Linear regulator.** A linear regulator (refer Fig. 1) acts as a variable resistor. The resistance of the regulator varies in accordance with the load, resulting in a constant output voltage.

The primary filter capacitor is placed at the input of the regulator to help filter-out the 50-cycle ripple. This

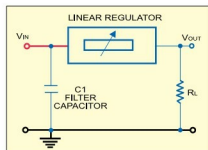


Fig. 1: Circuit of a linear regulator

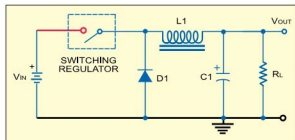


Fig. 2: Switching regulator circuit

arrangement has a serious drawback in terms of efficiency. For example, if the input voltage is 12V and output voltage 5V, the drop across the regulator will be 7V. If the load current is 2 amperes, then out of 24-watt power drawn from the source only 10 watts is delivered to the load, as 14 watts gets dissipated in the form of heat energy.

The more the voltage difference across the regulator, the lower the efficiency. A suitable heat-sink has to be used to prevent damage to the regulator due to excessive heat generated.

Switching regulators. There are three types of switching regulator circuits:

1. Buck circuit that produces a voltage less than the supply voltage
2. Boost circuit that produces a voltage greater than the supply voltage
3. Negative boost circuit that produces a voltage opposite in sign to the supply voltage

A linear regulator can only produce a voltage less than the supply voltage. But a switching regulator is much more flexible—very much like a DC analogue of the AC transformer.

The basic switching regulator circuit is shown in Fig. 2. It is a simple switch that goes on and off at a fixed rate, usually between 50 kHz and 100 kHz as set by the circuit. As is apparent, a switching regulator is much more efficient than a linear regulator. Diode (D1) and inductor (L1) are used in almost all switching regulators.

The diode has to be Schottky or some other very fast-switching diode. The inductor should use a core that does not saturate under high currents. The capacitor (C1) is normally low equivalent series resistance (ESR) type. When the switch is closed, an electromagnetic field is created around the inductor and the load connects to the source. The diode is reverse-biased.

When the switch is open, the inductor disconnects from the source and the current through it ceases. The electromagnetic field built up in inductor L1 now discharges, generating current in reverse polarity. As a result, diode D1 conducts until the field in L1 diminishes.

The time for which the switch remains closed during each switch-cycle is varied to maintain a constant output voltage. Because the switching system operates in 50-100kHz region and has an almost square waveform, it is rich in harmonics in high-frequency and even very-high-frequency/ultra-high-frequency regions. Special filtering is required, along with shielding, minimised lead-lengths and various toroidal filters

on leads going outside the case.

Switching regulator integrated circuits (ICs) are readily available. The LM2576 switching controller is one such IC. The chip includes a switch, voltage reference and feedback path for use in either the buck- or negative-boost circuits. It has a capacity of 3A, 40V maximum supply voltage (60V in the high-voltage LM2576HV version). Feedback resistors may be included for output voltages of 3.3, 5.0, 12 or 15V. The ADJ version uses external feedback resistors and can be adjusted to any output voltage.

Q2. What is the difference between personal simulation program with integrated circuit emphasis (PSPice) and very high-speed integrated circuit hardware description language (VHDL)?

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A2. PSPice, initially developed by MicroSim, is a SPICE analogue circuit and digital logic simulation program for Microsoft Windows used in electronic design automation. It has evolved into an analogue mixed-signal simulator. The software, now developed for more complex industry requirements, is integrated in the complete system design flow in OrCAD and Cadence Allegro.

SPICE can do several types of circuit analysis. The most important ones are:

1. Non-linear direct-current (DC) analysis by calculating the DC transfer curve
2. Non-linear transient analysis by calculating the voltage and current as a function of time when a large signal is applied
3. Linear alternating-current (AC) analysis by calculating the output as a function of frequency
4. Noise analysis
5. Sensitivity analysis
6. Distortion analysis
7. Fourier analysis by calculating and plotting the frequency spectrum
8. Monte Carlo analysis